

Aim: Write a JavaScript code for:

a. Performing various mathematical operations such as calculating factorial/ finding Fibonacci series/ displaying prime numbers in a given range/evaluating expressions/ calculating reverse of a number.

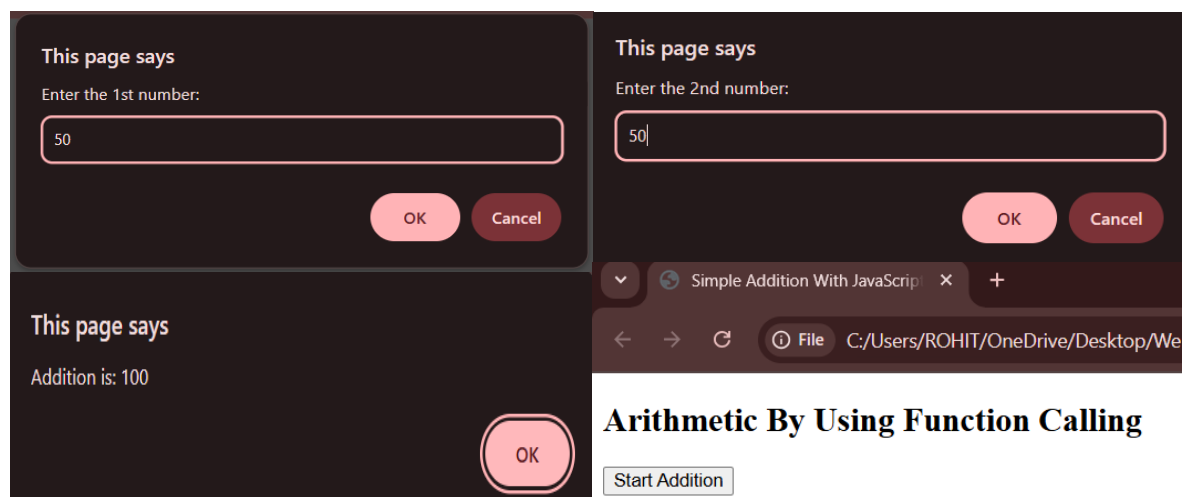
b. Validating the various form elements.

1. Simple arithmetic calculations.

CODE:

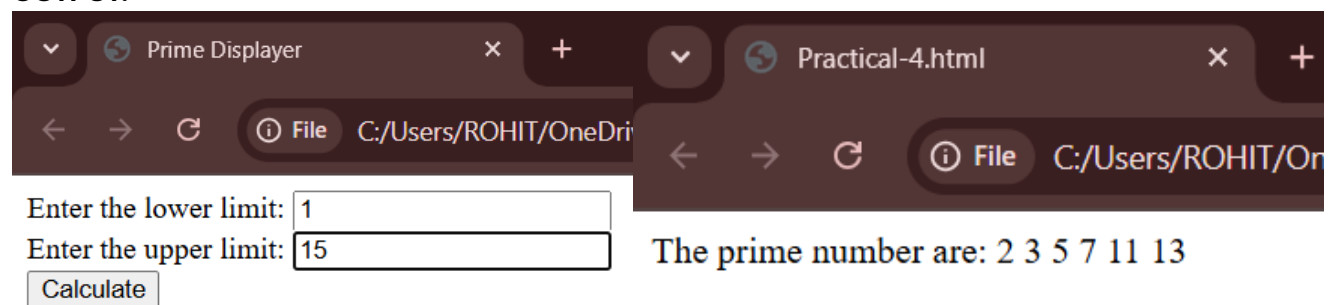
```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Addition With JavaScript</title>
</head>
<body>
  <h2>Arithmetic By Using Function Calling</h2>
  <button onclick="startAddition()">Start Addition</button>
  <script>
    function add(x, y) {
      return x + y;
    }
    function startAddition() {
      var x = Number(prompt("Enter the 1st number:"));
      var y = Number(prompt("Enter the 2nd number:"));
      var r = add(x, y);
      alert("Addition is: " + r);
    }
  </script>
</body>
</html>
```

OUTPUT:



2.Code to display prime number.**CODE:**

```
<!DOCTYPE html>
<html>
<head>
<title>Prime Displayer</title>
<script>
function prime(){
  var lw = Number(document.getElementById("n1").value);
  var up = Number(document.getElementById("n2").value);
  var i,j,c=0;
  document.write("The prime number are:");
  for(i=lw;i<=up;i++){
    for(j=2;j<i;j++){
      if(i%j==0){
        c=c+1;
      }
    }
    if(c==0 && i>1){
      document.write(" " + i);
    }
    c=0;
  }
}
</script>
</head>
<body>
  Enter the lower limit: <input type="text" id="n1"><br>
  Enter the upper limit: <input type="text" id="n2"><br>
  <button onclick="prime()">Calculate</button>
</body>
</html>
```

OUTPUT:

3.Code for Fibonacci Series.**CODE:**

```
<!DOCTYPE html>
<html>
<head>
<title>Fibonacci Series</title>
</head>
<body>
<script type = "text/javascript">
    var n1=0, n2=1, n3, i;
    var n = parseInt(prompt("Enter the limit for Fibonnaci Series:"));
    document.write("Fibonacci Series is ");
    for(i=1;i<=n;i++){
        document.write(" " + n1);
        n3 = n1 + n2;
        n1 = n2;
        n2 = n3;
    }
</script>
</body>
</html>
```

OUTPUT: